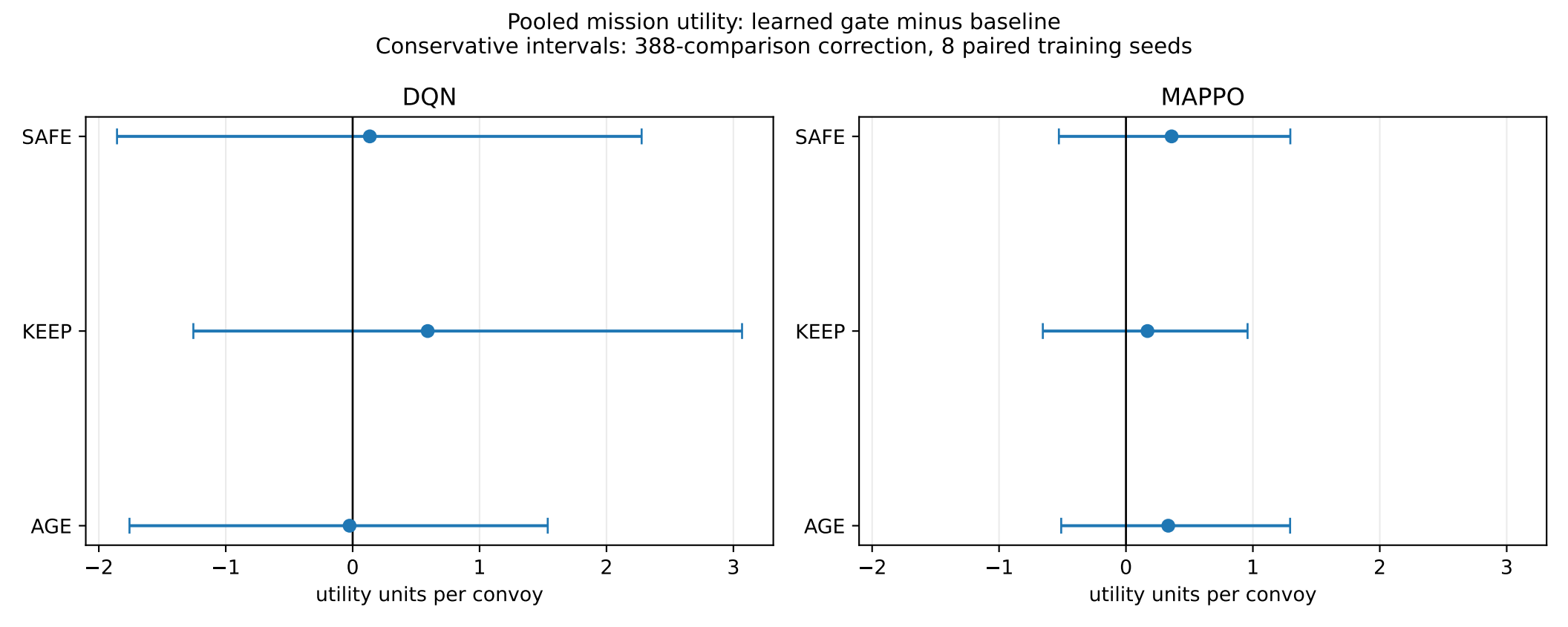
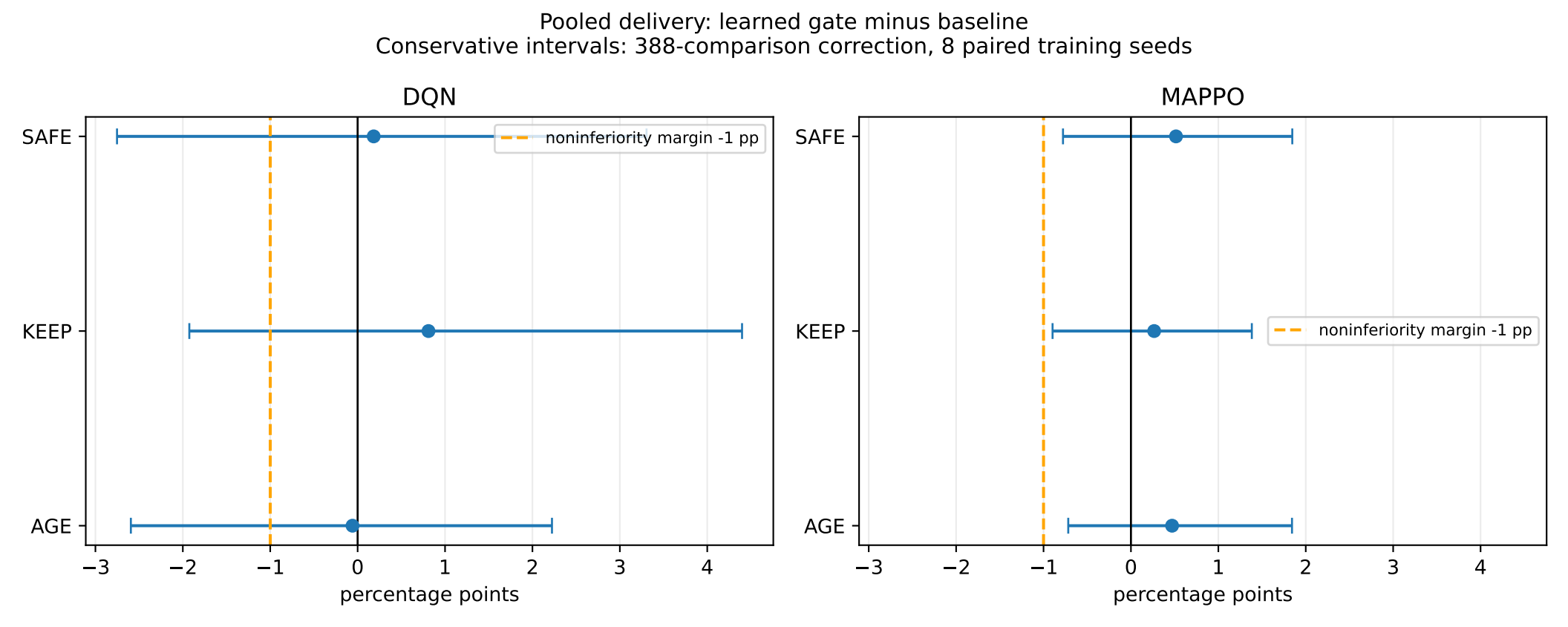
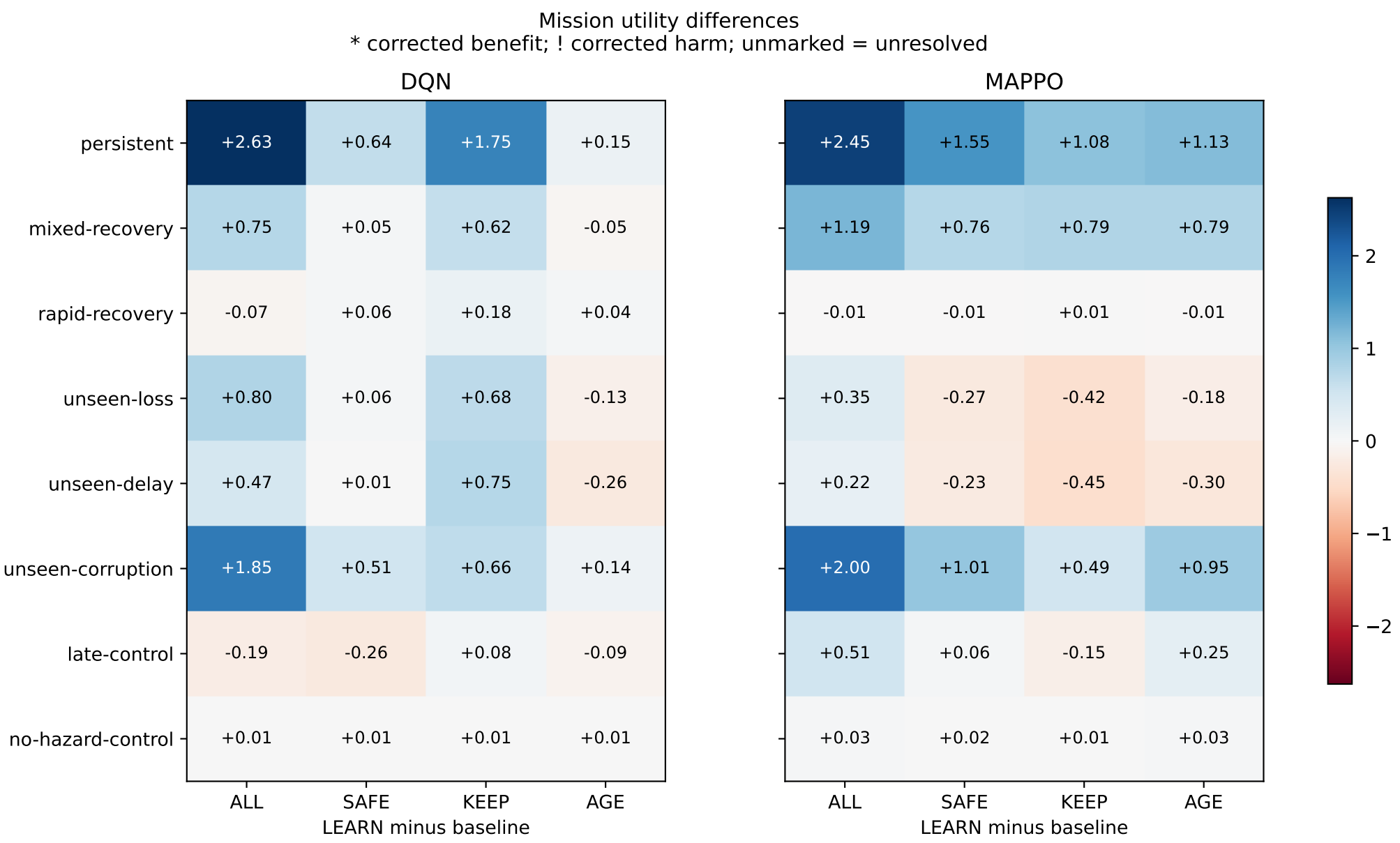
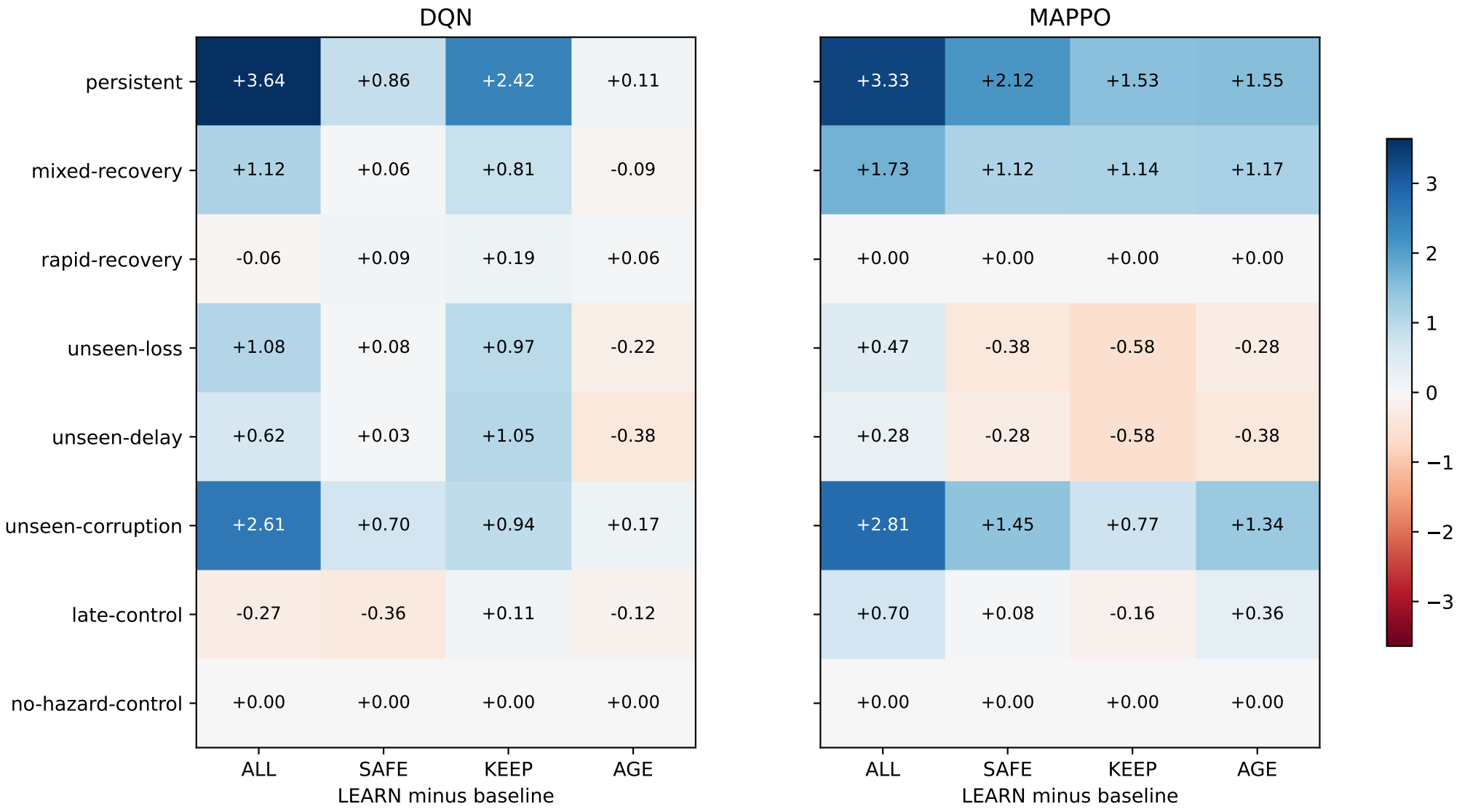
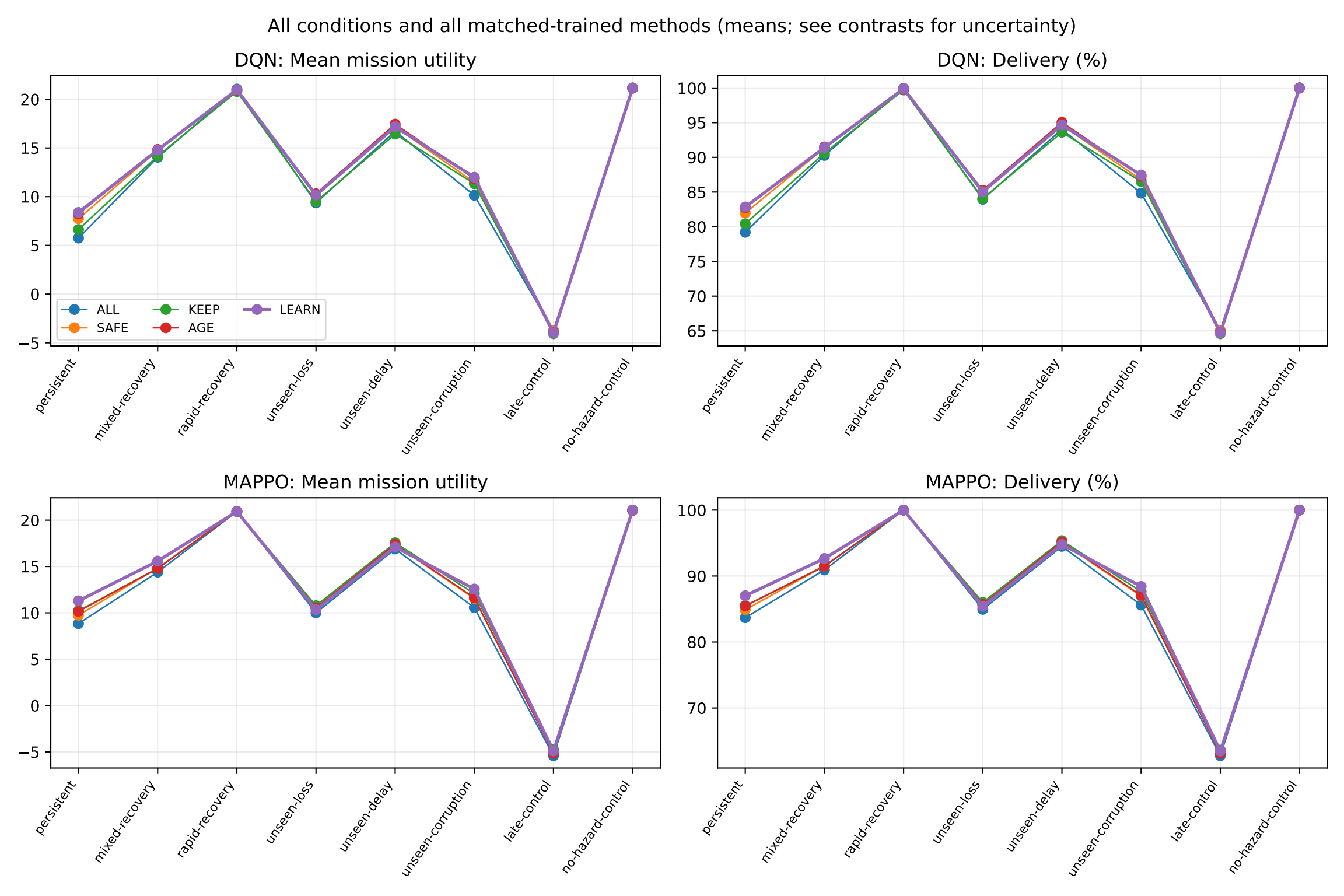
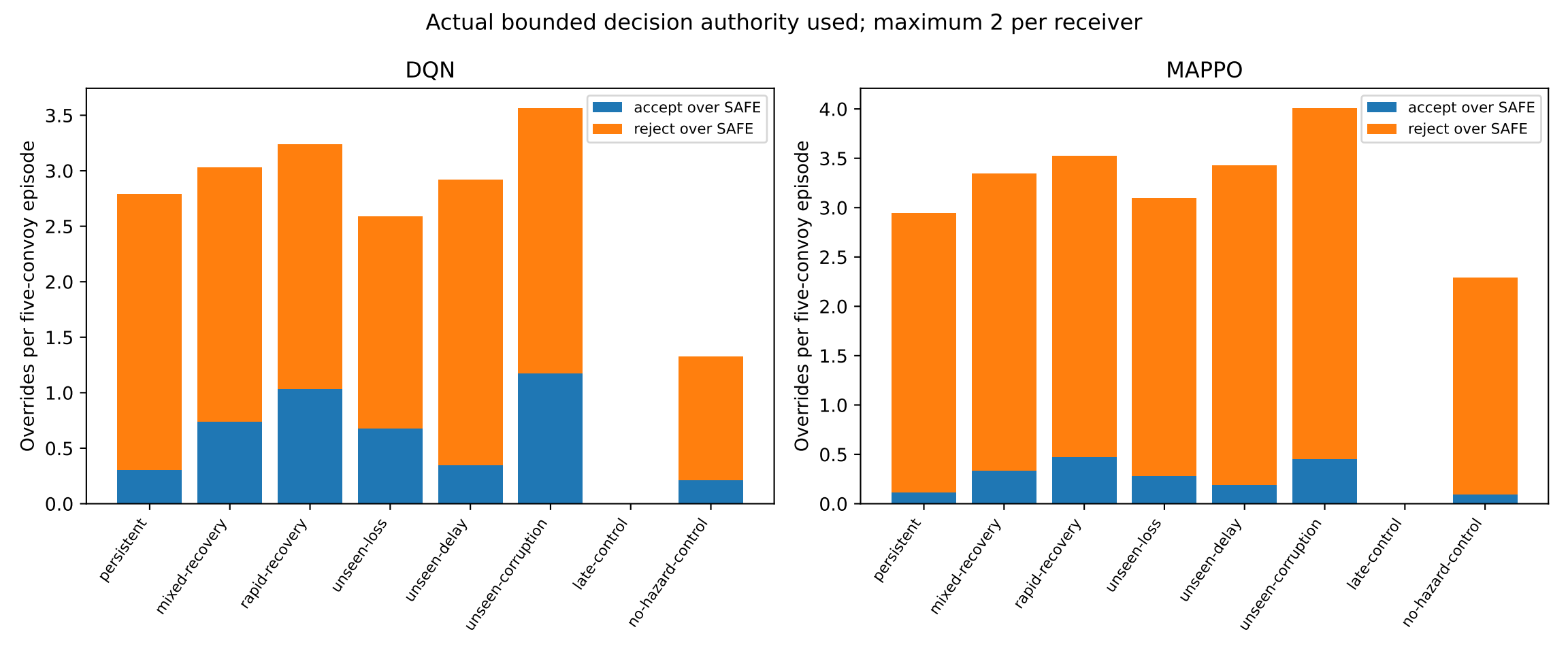

Pooled mission utility: learned gate minus baseline
Conservative intervals: 388-comparison correction, 8 paired training seeds
DQN
MAPPO
SAFE
KEEP
AGE
−2
−1
0
1
2
3
utility units per convoy

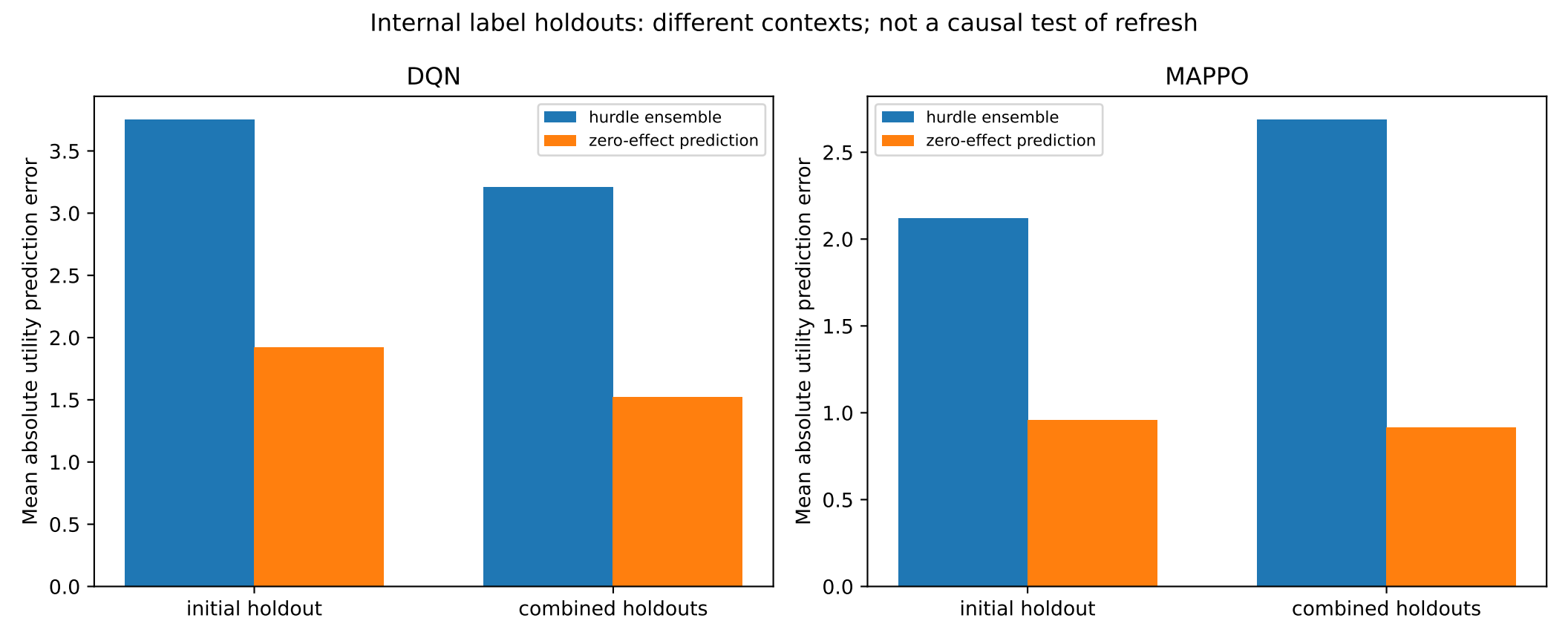

Pooled delivery: learned gate minus baseline
Conservative intervals: 388-comparison correction, 8 paired training seeds
DQN
MAPPO
SAFE
KEEP
AGE
noninferiority margin -1 pp
−3
−2
−1
0
1
2
3
4
percentage points

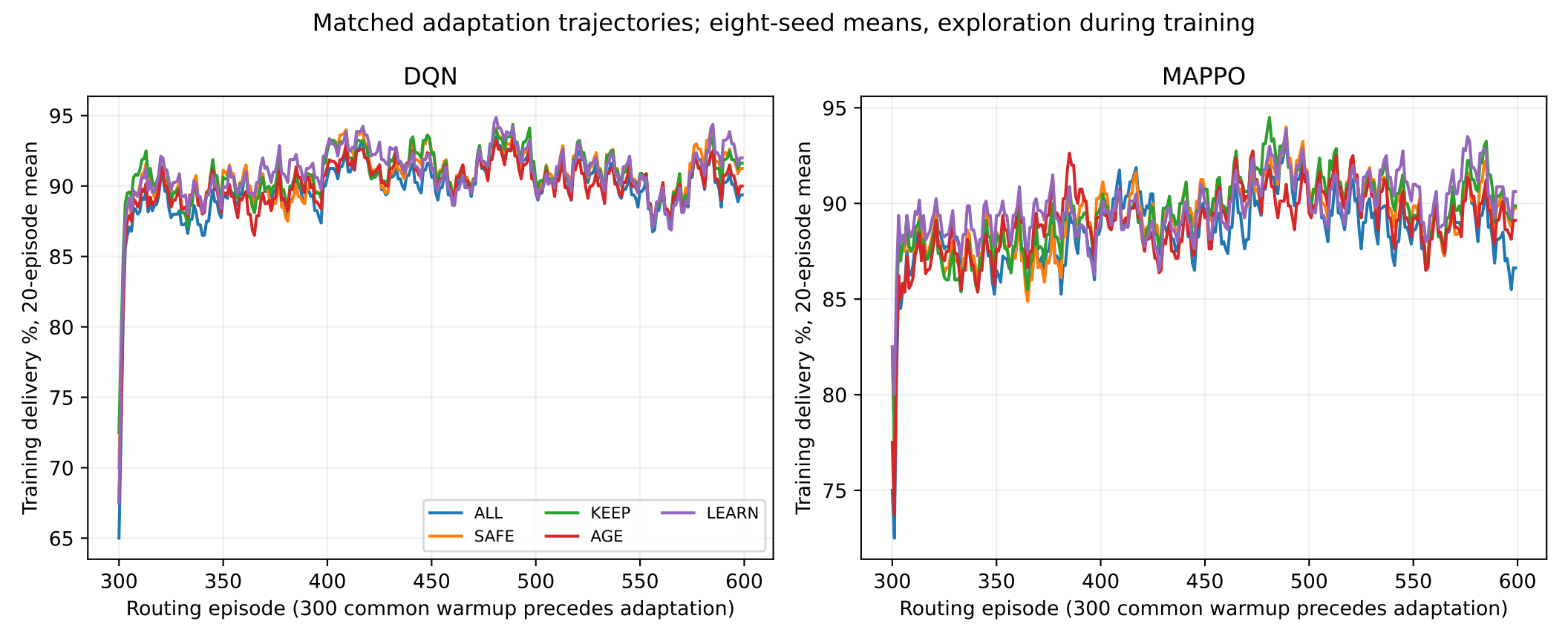

Mission utility differences
* corrected benefit; ! corrected harm; unmarked = unresolved
DQN
MAPPO
persistent
mixed-recovery
rapid-recovery
unseen-loss
unseen-delay
unseen-corruption
late-control
no-hazard-control
+2.63 +0.64 +1.75 +0.15
+0.75 +0.05 +0.62 -0.05
-0.07 +0.06 +0.18 +0.04
+0.80 +0.06 +0.68 -0.13
+0.47 +0.01 +0.75 -0.26
+1.85 +0.51 +0.66 +0.14
-0.19 -0.26 +0.08 -0.09
+0.01 +0.01 +0.01 +0.01
+2.45 +1.55 +1.08 +1.13
+1.19 +0.76 +0.79 +0.79
-0.01 -0.01 +0.01 -0.01
+0.35 -0.27 -0.42 -0.18
+0.22 -0.23 -0.45 -0.30
+2.00 +1.01 +0.49 +0.95
+0.51 +0.06 -0.15 +0.25
+0.03 +0.02 +0.01 +0.03
ALL SAFE KEEP AGE
LEARN minus baseline
ALL SAFE KEEP AGE
LEARN minus baseline
2
1
0
−1
−2


Delivery differences (percentage points)
* corrected benefit; ! corrected harm; unmarked = unresolved
DQN
MAPPO
persistent
mixed-recovery
rapid-recovery
unseen-loss
unseen-delay
unseen-corruption
late-control
no-hazard-control
+3.64 +0.86 +2.42 +0.11
+1.12 +0.06 +0.81 -0.09
-0.06 +0.09 +0.19 +0.06
+1.08 +0.08 +0.97 -0.22
+0.62 +0.03 +1.05 -0.38
+2.61 +0.70 +0.94 +0.17
-0.27 -0.36 +0.11 -0.12
+0.00 +0.00 +0.00 +0.00
+3.33 +2.12 +1.53 +1.55
+1.73 +1.12 +1.14 +1.17
+0.00 +0.00 +0.00 +0.00
+0.47 -0.38 -0.58 -0.28
+0.28 -0.28 -0.58 -0.38
+2.81 +1.45 +0.77 +1.34
+0.70 +0.08 -0.16 +0.36
+0.00 +0.00 +0.00 +0.00
ALL SAFE KEEP AGE
LEARN minus baseline
ALL SAFE KEEP AGE
LEARN minus baseline
3
2
1
0
−1
−2
−3


All conditions and all matched-trained methods (means; see contrasts for uncertainty)
DQN: Mean mission utility
DQN: Delivery (%)
MAPPO: Mean mission utility
MAPPO: Delivery (%)
ALL
SAFE
KEEP
AGE
LEARN
persistent
mixed-recovery
rapid-recovery
unseen-loss
unseen-delay
unseen-corruption
late-control
no-hazard-control


Actual bounded decision authority used; maximum 2 per receiver
DQN
MAPPO
Overrides per five-convoy episode
accept over SAFE
reject over SAFE
persistent
mixed-recovery
rapid-recovery
unseen-loss
unseen-delay
unseen-corruption
late-control
no-hazard-control


Internal label holdouts: different contexts; not a causal test of refresh
DQN
MAPPO
hurdle ensemble
zero-effect prediction
Mean absolute utility prediction error
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
initial holdout
combined holdouts


Matched adaptation trajectories; eight-seed means, exploration during training
DQN
MAPPO
Training delivery %, 20-episode mean
Routing episode (300 common warmup precedes adaptation)
ALL
SAFE
KEEP
AGE
LEARN
95
90
85
80
75
70
65
300
350
400
450
500
550
600